

# Complex Systems Exercise 5

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## 1 Dimensional Analysis

### 1.1 The pi theorem

Every physical law that relates  $n$  physical quantities containing  $k$  independent dimensions can be expressed as a relation between  $p = n - k$  dimensionless variables, which are often noted  $\pi$ .

### 1.2 What to do?

Steps (or comments? Or general things to do?):

- Write the dimension of each quantity.
- Find the smallest set of independent dimensions (default to SI, but attempt complex dimensions as well).
- Write down the dimensionless variables.
- Write one dimensionless variable as a function of the others. Don't forget to add a function for each dimensionless quantity.
- Get a self similar form: an equation for a dimensionless variable.
- Solve it and go back to initial variables.

## 2 Continuum Mechanics

### 2.1 Density, Velocity, and Compressability

Continuity equation

$$\partial_t \rho + \nabla \cdot (\rho v) = 0 \quad (1)$$

#### Incompressibility

Either  $\rho = \rho_0$  is a constant in space and time, or (the more common definition):

$$\nabla \cdot v = 0 \quad (2)$$

For an incompressible fluid (by the second definition), the Continuity equation becomes:

$$\partial_t \rho + v \cdot \nabla \rho = 0 \quad (3)$$

#### Lagrangian Coordinates (or material coordinates, or convected coordinates)

These are coordinates that move with the fluid. The material derivative is defined as:

$$D_t = \partial_t + v \cdot \nabla \quad (4)$$

Given a regular (Eulerian) coordinate  $x$ , then  $X$  is a Lagrangian Coordinate if:

$$D_t f = \left. \frac{\partial f}{\partial t} \right|_X \quad (5)$$

From the Continuity equation:

$$D_t \rho = -\rho \nabla \cdot v \quad (6)$$

So for an incompressible fluid the material derivative of  $\rho$  is zero.

### 2.2 Conservation of Momentum

$$D_t v = F + \frac{1}{\rho} \nabla \sigma \quad (7)$$

Where  $F$  is the force per unit mass and  $\sigma_{ij}$  is  $i$  component the force per unit area with normal in the  $j$ th direction.

**Flow by inertia** When  $F = 0$  and  $\sigma = 0$ :

$$D_t v = 0 \quad (8)$$

Solved (in 1D) by  $v(x, t)$  where:

$$v = f(\xi) \quad \xi = x - vt \quad f(x) = v(x, 0) \quad (9)$$

$$\partial_x v = \frac{f'}{1 + tf'} \quad \partial_t v = -\frac{vf'}{1 + tf'} \quad D_t \xi = 0 \quad (10)$$

For  $f' < 0$  there exists a time

$$t^* = \frac{1}{f'} = \frac{1}{v^*} \quad (11)$$

Where the velocity gradient  $\partial_x v$  diverges.

### 2.3 Ideal Incompressible Flow

For a fluid with isotropic pressure  $\sigma_{ij} = -\delta_{ij}p$ :

$$D_t v = F - \frac{1}{\rho} \nabla p \quad (12)$$

Assuming Incompressibility  $\nabla \cdot v = 0$  and conservative forces  $\nabla \times F = 0$ , and taking the curl gives:

$$\partial_t \omega + v \cdot \nabla \omega + \omega \cdot \nabla v = 0 \quad (13)$$

Where  $\omega = \nabla \times v$  is the **vorticity**. Lagrange's theorem states that if  $\omega(t=0) = 0$  then  $\omega = 0$  in all times.

**Potential flow** If  $\omega = 0$ , then  $v = \nabla \phi$  for some potential function  $\phi$  and compressability is  $\nabla^2 \phi = 0$ .

## 2.4 Bernoulli's Principle

Assuming Incompressibility and that forces are gravitational  $\mathbf{F} = -g\hat{\mathbf{z}}$ :

$$\frac{1}{2}\rho v^2 + P + \rho g z = \text{const} \quad (14)$$

If the  $\omega = 0$  (potential flow), this is true everywhere. If  $\omega \neq 0$ , this is true **along streamlines** (constant Lagrangian coordinates).

## 2.5 Viscous flow

Symmetry of the stress tensor:

$$\sigma_{ij} = \sigma_{ji} \quad (15)$$

Gives the **Navier Stokes equations**:

$$\rho D_t \mathbf{v} = \rho \partial_t \mathbf{v} + \rho (\mathbf{v} \cdot \nabla) \mathbf{v} = \rho \mathbf{F} - \nabla p + \eta \nabla^2 \mathbf{v} + (\mu + \eta) \nabla (\nabla \cdot \mathbf{v}) \quad (16)$$

Assuming incompressibility and naming kinematic viscosity  $\nu = \mu/\rho$ :

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \mathbf{F} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} \quad (17)$$

### Poiseuille flow

Steady flow  $\partial_t v = 0$ , of a viscous incompressible fluid  $(\rho, \eta)$ , in a round pipe with radius  $R$ , with no external forces, flowing only along the pipe  $\partial_\phi v = \partial_r v = 0$ , assuming constant pressure gradient in the flow direction  $\partial_x p = -G$ :

$$\nabla p = \eta \nabla^2 v_x = \frac{\eta}{r} \partial_r (r \partial_r v_x(r)) \quad (18)$$

Requiring no divergences at  $r = 0$  and  $v_x(R) = 0$ :

$$v_x = -\frac{\Delta p}{4\eta L} (R^2 - r^2) \quad (19)$$

And the mass flux:

$$Q = 2\pi \int_0^R v_x r dr = \frac{\pi R^4}{8\eta} \frac{-\Delta p}{L} \quad (20)$$

### Reynold number

Dimensionless number describing the relations in NS equation.

$$Re = \frac{lv}{\nu} = \frac{lv\rho}{\eta} \quad (21)$$

Where  $l, v$  are the typical length scale and velocity. In Navier-Stokes:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \mathbf{F} - \frac{1}{\rho} \nabla p + \frac{1}{Re} \nabla^2 \mathbf{v} \quad (22)$$

- $Re \gg 1$  - the system is dominated by inertia
- $Re \ll 1$  - inertia is negligible (Stokes flow)

### Stokes flow

Neglecting the inertia terms:

$$\mathbf{F} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} = 0 \quad (23)$$

So, for an object with length scale  $r_0$  (like a sphere), using dimensional analysis

$$F = C\eta r_0 v \quad (24)$$

For sphere,  $C = 6\pi$ .

### The Oseen Tensor

The Green function for Stokes flow equation. For a force distribution  $f(r)$ :

$$p(r) = \int g(r - r') f(r') d^3 r' \quad (25)$$

$$v(r) = \int T(r - r') f(r') d^3 r' \quad (26)$$

We get:

$$g_i = \frac{r_i - r'_i}{4\pi |\mathbf{r} - \mathbf{r}'|^3} \quad (27)$$

$$T_{ij} = \frac{1}{8\pi |\mathbf{r} - \mathbf{r}'|} \left( \delta_{ij} + \frac{(r_i - r'_i)(r_j - r'_j)}{|\mathbf{r} - \mathbf{r}'|^2} \right) \quad (28)$$

## 2.6 Hydrodynamics Instabilities

The instability of two fluid with relative tangential velocity along their surface x-y. Since in Navier Stokes for a compressible fluid, the pressure follows Laplace's equation  $\nabla^2 p = 0$ , and there can be pressure waves:

$$p = p_0 + \delta p \quad \delta p = A e^{i(kx - \omega t) - kz} \quad (29)$$

Allowing for velocity waves as well ( $v$  is the relative velocity of  $\rho_1$  with respect to  $\rho_2$ ):

$$v = v_0 + \delta v \quad \delta v = B e^{i(kx - \omega t) - kz} \quad (30)$$

solving Navier-Stokes, and requiring that the fluids interface  $z = \zeta(x, t)$  has the same velocity gives:

$$\omega = kv_0 \frac{\alpha \pm i\sqrt{\alpha}}{1 + \alpha} \quad \alpha = \frac{\rho_1}{\rho_2} \quad (31)$$

Which has an imaginary part, marking instability.

### Rayleigh Plateau instability

The instability of a surface given a wave in it. In a cylinder of fluid with radius (with a volume preserving perturbation):

$$r = r_0 + \epsilon \cos(kz) - \frac{\epsilon^2}{4\pi} \quad (32)$$

Will have an area:

$$A = \frac{4\pi^2}{k} \left( r_0 + \frac{\epsilon^2}{4\pi r_0} (r_0^2 k^2 - 1) \right) \quad (33)$$

Given the surface tension energy  $E = \gamma A$ , the energy is negative for  $k < 1/r_0$ , meaning that the perturbation will grow.

## 2.7 Surfaces

## 2.8 Surface Tension

$$E = \gamma A = \int \sqrt{|a|} dx^1 dx^2 \quad (34)$$

For an open surface, looking for minimum to the surface:

$$H = 0 \quad -\frac{1}{2\nabla^2}\eta + \eta K > 0 \quad \forall \eta \quad (35)$$

For a colsed surface there can be a finite pressure difference  $\delta p$ , using **Young-Laplace law**.

$$\Delta p = -2\gamma H \quad (36)$$

## 3 Elasticity

The momentum equation (similar to NS):

$$\partial_t v_i + v_j \partial_j v_i = f_i + \frac{1}{\rho} \partial_j \sigma_{ij} \quad (37)$$

For a static system  $v_i = 0$ :

$$\partial_j \sigma_{ij} = 0 \quad (38)$$

The linearized strain tensor for some displacement  $\mathbf{u} = \mathbf{r} - \mathbf{x}$ , where at  $\mathbf{x}$  the system is at rest:

$$\epsilon_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i) \quad (39)$$

Their relation is:

$$\sigma_{ij} = \lambda \delta_{ij} \text{Tr}(\epsilon) + 2\mu \epsilon_{ij} \quad (40)$$

Where  $\lambda, \mu$  are the **Lame coefficients**.

**Young's modulus and Poisson ratio**

$$\lambda = \frac{Y\nu}{(1+\nu)(1-2\nu)} \quad \mu = \frac{Y}{2(1+\nu)} \quad (41)$$

Young's modulue has units of pressure, and Poisson ratio is unitless, between  $-1$  and  $\frac{1}{2}$ .

### 3.1 Compression waves in solids

Length wave  $\epsilon_{xx} = \partial_x u_x$ , gives

$$\sigma_{ij} = (\lambda \delta_{ij} + 2\mu \delta_{ix} \delta_{jx}) \partial_x u_x \quad (42)$$

$$\partial_t^2 u_x = C_p^2 \partial^2 u_x \quad C_p^2 = \frac{Y}{\rho} \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \quad (43)$$

Trasverse wave  $\epsilon_{xy} = \partial_x u_y$ , gives

$$\sigma_{xy} = \sigma_{yx} = \mu \partial_x u_y \quad (44)$$

$$\partial_t^2 u_y = C_s^2 \partial^2 u_y \quad C_s^2 = \frac{Y}{\rho} \frac{1}{2(1+\nu)} \quad (45)$$

### 3.2 Large displacements

Strain to second order:

$$\epsilon_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i + \partial_i u_j \partial_j u_i) \quad (46)$$

Cauchy-Green strain:

$$\epsilon = \frac{1}{2} (g - \bar{g}) \quad (47)$$

Where  $g$  is the metric and  $\bar{g}$  is the original metric.

## 3.3 This structure elasticity

The 3D elastic energy functional:

$$E = \int \frac{1}{2} \sigma^{ij} \epsilon_{ij} d^3 x \quad (48)$$

For thin structure  $t, W \ll L$ :

$$E = YW \int \frac{1}{2} (\epsilon_1^1)^2 d^2 x \quad (49)$$

Thin configuration along a curve:

$$\mathbf{r}(u, v) = \boldsymbol{\gamma}(u) + v \hat{\mathbf{n}}(u) \quad \partial_u \boldsymbol{\gamma} = \frac{\partial s}{\partial u} \hat{\mathbf{t}} \quad (50)$$

Where  $\hat{\mathbf{t}}$  is the tangent and  $\hat{\mathbf{n}}$  is the normal:

$$\partial_u \hat{\mathbf{t}} = \frac{\partial s}{\partial u} \kappa \hat{\mathbf{n}} \quad \partial_u \hat{\mathbf{n}} = -\frac{\partial s}{\partial u} \kappa \hat{\mathbf{t}} \quad (51)$$

Assuming a small displacement  $\frac{\partial s}{\partial u} = 1 + \Delta$ , the metric is:

$$a = \begin{pmatrix} (1 - \nu k)^2 (1 + \Delta)^2 & \\ & 1 \end{pmatrix} \quad (52)$$

**The Kirchoff-Love assumptions:**

From the metric we get  $\epsilon_{22} = 0$ , and before we assumes  $\sigma_{22} = 0$ . These contradict. This finally gives:

$$E = \frac{YWt}{2} \int_0^L \Delta^2 du + \frac{YWt^3}{24} \int_0^L \kappa^2 du \equiv E_{stretch} + E_{bending} \quad (53)$$

## 4 Random Walks

**In 1D**

$$x_{n+1} = x_n \pm 1 \quad (54)$$

After  $N$  steps,  $R$  steps to the right and  $N_R$  steps to the left:

$$x_N = 2R - N \quad (55)$$

$$\langle x_N \rangle = 0 \quad \langle x_N^2 \rangle = N \quad (56)$$

$$p(x) = \begin{cases} 0 & x \text{ odd} \\ \frac{1}{2^N} \binom{N}{R} & x \text{ even} \end{cases} \quad (57)$$

Using Stirling's formula:

$$p_{even}(x) = \frac{2}{\sqrt{2\pi N}} e^{-\frac{x^2}{2N}} \quad (58)$$

**In higher dimensions**

Given  $p(\vec{r}, t)$ , we can write (time is still discrete, just with time steps  $\tau$ ):

$$p(\vec{r}, t + \tau) = \int p(\vec{r} - \vec{\Delta}t, t) g(\vec{\Delta}t) d^3 \vec{\Delta}t \quad (59)$$

Assuming isotropic distirbution and definint  $\partial_t p = \frac{1}{\tau} (p(t + \tau) - p(t))$ :

$$\partial_t p = D \nabla^2 p \quad D = \frac{1}{6d\tau} \langle \Delta^2 \rangle \quad (60)$$

This is the diffusion equation, with Gaussian Solutions:

$$p(\vec{r}, t) = \frac{1}{(4\pi Dt)^{3/2}} e^{-\frac{r^2}{4Dt}} \quad (61)$$

$$\langle r^2 \rangle = 6Dt \quad (62)$$

## 5 Markov Chains

Given the probability matrix  $P_{ij}$ : the probability to go from state  $i$  to state  $j$ , then the probability to be in state  $i$  at  $n + 1$  is:

$$p_i^{n+1} = \sum_j P_{ji} p_j \quad (63)$$

So:

$$\vec{p}_n = P^n \vec{p}_0 \quad (64)$$

## 6 Langevin and Fokker-Planck Equations

**Langevin equations** are the equations of motion with noise, and **Fokker-Planck equations** are the equations for the probability distribution.

For the simple equation

$$\dot{x} = \xi(t) \quad (65)$$

With  $\langle \xi(t)\xi(t') \rangle = \frac{D}{2}\delta(t - t')$ , then:

$$\langle x^2 \rangle = 2Dt \quad (66)$$

For the general Langevin equation:

$$m \frac{dv}{dt} = -\gamma v + \xi(t) - V'(x) \quad (67)$$

In the overdamped limit (removing the LHS and dividing by  $\gamma$ ):

$$\frac{dx}{dt} = \mu f(t) + \xi(t) \quad (68)$$

We get two currents - force current and diffusion current:

$$\vec{j}_f = -\mu p \vec{\nabla} V(\vec{r}) \quad \vec{j}_d = -D \vec{\nabla} p \quad (69)$$

So from the Continuity equation:

$$\partial_t p = -\vec{\nabla} \cdot [\vec{j}_f + \vec{j}_d] \quad (70)$$

## 7 Generalized CLT

Characteristic function:

$$\varphi(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt \quad (71)$$

For sum of variables  $X = \sum_i x_i$ :

$$\varphi_X(\omega) = \prod_i \varphi_i(\omega) \quad (72)$$

So:

$$p_n = \mathcal{F}^{-1}[\phi_n] = \mathcal{F}^{-1}[(\mathcal{F}[p])^n] \quad (73)$$

For a distribution whose tail goes as  $p(x > x^*) \propto x^{-(1+\mu)}$ :

- $1 < \mu < 2$ : the mean is defined but the variance isn't.
- $0 < \mu < 1$ : the mean is not defined.

**Levi stable distribution** is a distribution  $f(x)$  such that the sum of two random variables from the distribution has the same distribution. The most general such distribution has the characteristic function:

$$\varphi(\omega) = \exp[A|\omega|^\mu] \quad (74)$$

## 8 Extreme Value Theorem

The Cumulative Probability Function (CPF) of a maximum value scales multiplicatively:

$$G(x) = p(X_n < x) = \prod_i p(x_i < x) \quad (75)$$

For a distribution normalized using

$$\xi_n = \frac{x_n - b_n}{a_n} \quad (76)$$

Where

$$\alpha = \lim_{n \rightarrow \infty} \frac{n}{a_n} \frac{\partial a_n}{\partial n} \quad \beta = \lim_{n \rightarrow \infty} \frac{n}{a_n} \frac{\partial b_n}{\partial n} \quad (77)$$

Then:

- $\alpha > 0$  **Fréchet Distribution** has  $G(x) = e^{-ax^{-1/\alpha}}$
- $\alpha = 0$  **Gumbel Distribution** has  $G(x) = e^{-e^{-ax+b}}$
- $\alpha < 0$  **Weibull Distribution** has  $G(x) = e^{\alpha(\beta/|\alpha|-x)^{1/|\alpha|}}$

## 9 Random Matrix Theory

There are three meaningful ensembles:

- **Gaussian Orthogonal Ensemble**: matrices are real and symmetric, with probability  $p(A) \propto e^{-\frac{N}{4} \text{tr } A^2}$ .
- **Gaussian Unitary Ensemble**: matrices are complex and hermitian, with probability  $p(A) \propto e^{-\frac{N}{2} \text{tr } A^2}$ .
- **Gaussian Symplectic Ensemble**: matrices are complex and harmitian, with probability  $p(a) \propto e^{-n \text{tr } A^2}$

**Wigner's Surmise** says that the result of  $N = 2$  matrix are a good approximation to the  $n \gg 1$  case.

## 10 Dynamical Systems

### 10.1 Discrete maps

$$\vec{x}_{n+1} = f(\vec{x}_n) \quad (78)$$

Fixed point:

$$\vec{x}^* = f(\vec{x}^*) \quad (79)$$

GRaphically these are the intersection of the map and the  $y = x$  line.

Periodic orbits:

$$\vec{x}^* = f^n(\vec{x}^*) \quad (80)$$

Stability of a p-orbit is determined by its **Stability coefficient**:

$$\lambda_p = \prod_i^p f'(x_i) \quad \begin{array}{ll} |\lambda_p| > 1 & \text{unstable} \\ |\lambda_p| < 1 & \text{stable} \\ |\lambda_p| = 1 & \text{superstable} \end{array} \quad (81)$$

Bifurcations:

- Target bifurcation: two points, stable and unstable, appear/annihilate when the map  $f(x_n)$  becomes tangent to  $y = x$ . Includes a slowing of the dynamical system before the creation (Intermittency).
- Period Doubling bifurcation: a stable fixed point becomes unstable and creates two stable 2-orbits. This happens when the position of the fixed point is not changing, but  $f'(x^*)$  crosses 1 or  $-1$ . When crossing 1, this is very similar to 1D continuous system Pitchfork bifurcation. When crossing  $-1$ , then  $f^{2'}(x^*)$  crosses 1, and gets two stable fixed point of  $f^2$  are created. this essentially doubles the orbit of the 1-orbit to 2-orbit, and if  $f$  is actually  $f^p$ , then it doubles the p-orbit to 2p-orbit.
- Inverse Period Doubling bifurcation: the same as period doubling, but a single unstable fixed-point (p-orbit) becomes two unstable 2-orbits (2p-orbits) and a stable fixed point.

### Sarkovskii's theorem

Given Sarkovskii's order:

$$3, 5, 7, \dots, 2 \cdot 3, 2 \cdot 5, \dots, 2^2 \cdot 3, 2^2 \cdot 5, \dots, 2^3, 2^2, 2, 1 \quad (82)$$

Then if a map has p-periodic orbit, it has periodic orbits of all order above it.

## 10.2 Continuous time

$$\dot{\vec{x}} = f(t, \vec{x}) \quad (83)$$

Fixed point:

$$f(\vec{x}) = 0 \quad (84)$$

### 10.2.1 1D systems

Categorization of fixed points:

- $f'(x^*) < 0 \rightarrow$  stable
- $f'(x^*) > 0 \rightarrow$  unstable
- $f'(x^*) = 0 \rightarrow$  look at higher derivatives. if all derivatives are zero, the point is half-stable.

The characteristic time scale:

$$\tau = |f'(x^*)|^{-1} \quad (85)$$

### Bifurcations

- Saddle-Node bifurcation ( $r + x^2$ ): when two fixed points join to become a half stable fixed point, then vanish.
- Transcritical bifurcation ( $rx + x^2$ ): when a node changes its stability.
- Pitchfork bifurcation - in a symmetric system, a fixed point is split into three - one unstable and two stable fixed points.

These can be **supercritical** ( $rx - x^3$ ), where the  $x^3$  term causes the decay to be algebraic with time (critical slowdown) or **subcritical** ( $rx + x^3 - x^5$ ), where there are additional two stable fixed points, created by two simultaneous saddle-point bifurcations, and  $x = 0$  is only locally stable.

## 10.2.2 2D systems

### Stability

- Attracting fixed point: points in the neighborhood go to the fixed point at  $t \rightarrow \infty$ .
- Lyapunov stability - If a point in the neighborhood remains in the neighborhood for all times  $t$ .
- A point that has stability in one direction and instability in the other is called a saddlepoint. It has a stable manifold (all points that flow to the fixed point at  $t \rightarrow \infty$ ) and unstable manifold (the same at  $t \rightarrow -\infty$ ).

### Linear system

$$\dot{\vec{x}} = A\vec{x} \quad (86)$$

So  $\vec{x} = (0, 0)$  is always a fixed point. Given that  $\lambda_{1,2}$  are the eigenvalues of  $A$  and  $\vec{v}_{1,2}$  are the eigenvectors, then:

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \quad (87)$$

Where  $\tau$  is the trace and  $\Delta$  is the determinant, and the general solution starting from  $\vec{x}(0) = c_1\vec{v}_1 + c_2\vec{v}_2$  is:

$$\vec{x}(t) = c_1\vec{v}_1e^{\lambda_1 t} + c_2\vec{v}_2e^{\lambda_2 t} \quad (88)$$

Then the point can be written as:

- $\Delta < 0$  - saddlepoint: one eigenvalue is negative and one is positive.
- $\Delta > 0$  and  $\tau^2 > 4\Delta$  - node: then the eigenvalues have the same sign. If  $\tau > 0$  then they are both positive, and the node is unstable. If  $\tau < 0$  then the point is stable.
- $\Delta > 0$  and  $\tau^2 < 4\Delta$  - spirals: the eigenvalues are complex, and based on the sign of  $\tau$  can have either stability.
- $\Delta > 0$  and  $\tau = 0$ : centers - the eigenvalues are purely imaginary.
- $\tau^2 = 4\Delta$ : the eigenvalues are the same, and we get a star (with stability based on  $\tau$ ) if the multiplicity is algebraic and a degenerate node, with a single direction, if the multiplicity is geometric.

**Nonlinear system** Given:

$$\dot{x} = f(x, y) \quad \dot{y} = g(x, y) \quad (89)$$

We linearize around a fixed point:

$$\dot{x} = Jx \quad J = \begin{pmatrix} \partial_x f & \partial_y f \\ \partial_x g & \partial_y g \end{pmatrix} \quad (90)$$

If we get a marginal case - any line in the  $\tau, \Delta$  graph, be cautious!

A **hyperbolic** point is a point where  $\text{Re}(\lambda) \neq 0$  for all eigenvalues. **The Hartman-Grobman theorem** states that locally, the phase near hyperbolic points are "topologically equivalent" to the linearized version.

Centers are nevertheless stable in **conservative systems** (systems with a conserved quantity, i.e. energy), and systems with **time reversal symmetry**.

### Index Theory

For a system  $\vec{x} = \vec{f}(\vec{x})$  we define:

$$\phi(x) = \text{atan}\left(\frac{\dot{y}}{\dot{x}}\right) \quad (91)$$

Going along a closed loop  $C$  in  $\mathbb{R}^2$ , the loop index is defined as the number of cycles  $\phi$  does:

$$I_C = \frac{1}{2\pi} \oint \phi(x(\lambda)) d\lambda = \frac{1}{2\pi} \oint \left( \frac{\ddot{x}}{\dot{x}} - \frac{\ddot{y}}{\dot{y}} \right) d\lambda \quad (92)$$

- The loop always goes counterclockwise. The index doesn't depend on the direction of time.
- The contour can be smoothly deformed, without going through any fixed points, and not change its index.
- If  $C$  is a trajectory, then  $I_C = 1$ .
- The index is 0 if there are no fixed points inside the loop.
- This means that index of an isolated fixed point is the index of any loop that contains only itself.
- The index of a fixed point is independent of its stability.
- The index of nodes, degenerate nodes, spirals and centers is 1, and of saddles is  $-1$ .
- If a closed curve  $C$  encloses multiple fixed points, its index is the sum of their indices:
- Any closed orbit in state-space has an index  $IC = +1$ , and therefore it must enclose fixed points whose indices sum up to  $+1$ .

### Limit Cycles

An isolated closed trajectory.

There are no limit cycles:

- In potential systems  $\vec{x} = -\vec{\nabla}V$ .
- If there is a Liapunov Function: a function  $V(x)$  such that at the fixed point  $V(x^*) = 0$  and everywhere else  $V(x) > 0$  and  $\dot{V}(x) < 0$ .
- Dulac's Criterion: In an area where  $\nabla \cdot (g\vec{f})$  doesn't change sign, for every  $g(x)$ .

**Poincare-Bendixson Theorem** Consider a close bounded subset of the plane  $R$ , and that the vector field  $\vec{f}(\vec{x})$  is smooth enough in some open subset of  $R$ . If  $R$  does not have a fixed point, and if at least one trajectory never leaves  $R$ , then  $R$  contains a closed orbit.

To use this, we find a **attracting zone**: we plot the null-cline and find an area where the normal derivative to the boundary is always inward.

**Bifurcations** Has all the 1D bifurcations, and also **Hopf bifurcations**.

- When the eigenvalues don't have an imaginary part, it's the same as in 1D.
- If they have an imaginary part, and they change sign of the real value, it's a **supercritical Hopf bifurcation** - moving from stable spiral to an unstable spiral, creating a limit cycle.
- The **subcritical Hopf bifurcation** similar to 1D subcritical pitchfork, with term  $x^3$  unstabilizes and an additional  $x^5$  stabilizing.
- It is impossible to tell the type of Hopf bifurcation from linearization.

## 11 Math

### 11.1 Tensors

Symmetric rank two tensors in 3D, linear to a generic tensor  $A_{ij}$ :

$$\sigma_{ij} = \alpha \delta_{ij} + \beta \text{Tr}(A) \delta_{ij} + \gamma (A_{ij} + A_{ji}) \quad (93)$$

Symmetric, uniform, isotropic, rank four tensor:

$$A^{ijkl} = \lambda \delta^{ij} \delta^{kl} + \mu (\delta^{ik} \delta^{jl} + \delta^{il} \delta^{jk}) \quad (94)$$

**Isotropy** When requiring isotropy in a vector relation:

- Write the relation in tensor form  $C^i = T^{ijk} A_j B_k$
- Move to another system  $\Lambda_j^i$ :

$$C'^i = T^{jkl} \Lambda_j^i \Lambda_k^n \Lambda_l^m A_n B_m \quad (95)$$

- Require  $T$  to be invariant

$$T^{inm} = T^{jkl} \Lambda_j^i \Lambda_k^n \Lambda_l^m \quad (96)$$

- Rotate by different degrees to see relations between the components of  $T$

In the case of the bilinear tensor relation  $C^i = T^{ijk} A_j B_k$ :

$$T^{ijk} \propto \epsilon^{ijk} \quad (97)$$

### 11.2 Surfaces

Given a 2D manifold in 3D space:

$$\mathbf{r}(x^1, x^2) : \mathcal{D} \rightarrow \mathcal{S} \quad (98)$$

Where  $\mathcal{D} \in \mathbb{R}^2$  and  $\mathcal{S} \in \mathbb{R}^3$ .

Tangent vectors:

$$\partial_1 \mathbf{r}, \partial_2 \mathbf{r} \quad (99)$$

Distances are measured using the metric (first fundamental form):

$$ds^2 = g_{\alpha\beta} dx^\alpha dx^\beta \quad g_{\alpha\beta} = \partial_\alpha \mathbf{r} \cdot \partial_\beta \mathbf{r} \quad (100)$$

The normal vector:

$$\hat{\mathbf{N}} = \frac{\partial \mathbf{r} \times \partial_2 \mathbf{r}}{\sqrt{a}} \quad (101)$$

The second fundamental form (the curvature form):

$$b_{\alpha\beta} = \partial_\alpha \partial_\beta r \cdot \hat{\mathbf{N}} \quad (102)$$

The shape tensor:

$$s_\beta^\alpha = a^{\alpha\gamma} b_{\gamma\beta} \quad (103)$$

The physical normal curvature of curve  $(dx^1, dx^2)$

$$k_n(dx^1, dx^2) = \frac{dx^\alpha b_{\alpha\beta} dx^\beta}{dx^\alpha a_{\alpha\beta} dx^\beta} \quad (104)$$

For every given point, changing coordintes such that  $a$  is the identity, then rotating such that  $b$  is diagonal, gives:

$$b = \begin{pmatrix} \kappa_1 & \\ & \kappa_2 \end{pmatrix} \quad (105)$$

Where  $\kappa_1, \kappa_2$  are the **principal cuvatures**. They give the mean curvature and Gaussian curvature:

$$H = \frac{\kappa_1 + \kappa_2}{2} = \frac{1}{2} Tr(s) \quad K = \kappa_1 \kappa_2 = |s| = \frac{|b|}{|a|} \quad (106)$$

To get a minimal sruface:

$$H = 0 \quad -\frac{1}{2\nabla^2} \eta + \eta K > 0 \quad \forall \eta \quad (107)$$

### 11.3 Probabilities

Stirling's formula:

$$N! = \sqrt{2\pi N} \left( \frac{N}{e} \right)^N \quad (108)$$